

Glucosamine vs Collagen for Arthritis: Which Supplement Is B

Head-to-head comparison of glucosamine and collagen supplements for arthritis. Evidence-based analysis of effectiveness, dosage, safety and value for UK patients.

By **Living With Arthritis Editorial Team** · Pending clinical review ·
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Glucosamine and collagen are the two most popular joint supplements sold in the UK, with combined sales exceeding **£200 million annually**. But which one should you choose for arthritis? Is one better than the other? Can you take both? This evidence-based guide compares glucosamine and collagen head-to-head, examining the science, practical considerations and value for money.

Glucosamine: The Established Choice

What Is It?

Glucosamine is a naturally occurring compound found in cartilage — the tough tissue that cushions joints. Supplements are manufactured from shellfish shells (glucosamine sulphate or hydrochloride) or produced synthetically. It has been the most widely used joint supplement worldwide for over 30 years.

How It Works

Glucosamine is thought to provide building blocks for cartilage repair and maintenance, stimulate the production of proteoglycans (which give cartilage its shock-absorbing properties), have mild anti-inflammatory effects, and slow cartilage breakdown (though this is debated).

What the Evidence Says

The evidence for glucosamine is extensive but **mixed**. Positive findings include the GAIT study (the largest glucosamine trial with 1,583 participants) finding that glucosamine hydrochloride combined with chondroitin provided significant pain relief for moderate-to-severe knee OA

(though not for mild OA), European studies using glucosamine sulphate (Rotta brand) showing modest benefits for knee OA pain and function over 3 years, and a Cochrane review concluding that glucosamine provided "some degree of pain relief" compared to placebo. Negative findings include the large NIH-funded MOVES trial finding glucosamine+chondroitin no better than celecoxib (an NSAID), some meta-analyses showing benefits only when industry-funded studies are included, and the NHS and NICE not recommending glucosamine due to inconsistent evidence. The form matters — glucosamine **sulphate** has more positive evidence than glucosamine **hydrochloride**. The pharmaceutical-grade crystalline form (as used in the Rotta studies) may be more effective than over-the-counter preparations.

Dosage

Standard dose: 1,500 mg glucosamine sulphate daily (usually taken as a single dose or divided). Allow 8–12 weeks for potential effects.

Safety

Generally well tolerated. Side effects include mild digestive upset. Caution if you have a shellfish allergy (synthetic forms available). May affect blood sugar — discuss with your GP if diabetic. Possible interaction with warfarin (may increase anticoagulant effect).

Collagen: The Emerging Contender



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What Is It?

Collagen is the most abundant protein in the body, forming the structural framework of cartilage, bone, skin, tendons and ligaments. Supplements use hydrolysed collagen (broken down into smaller peptides for better absorption). Types include Type II collagen (native/undenatured, from chicken sternum — works through immune modulation), Type I and III hydrolysed collagen (from bovine, marine or porcine sources — provides building blocks), and collagen peptides (highly processed for maximum absorption).

How It Works

Collagen supplements are thought to provide amino acids (glycine, proline, hydroxyproline) needed for cartilage repair, stimulate the body's own collagen production in joint tissues, have anti-inflammatory effects through immune modulation (particularly Type II), and support surrounding structures (tendons, ligaments) as well as cartilage.

What the Evidence Says

The evidence for collagen is **promising but less extensive** than glucosamine. Positive findings include a meta-analysis of 5 RCTs finding that collagen supplementation significantly improved OA symptoms (pain, stiffness, function), a 24-week study of 10g daily hydrolysed collagen showing significant improvement in joint comfort during activity in athletes and active adults, Type II undenatured collagen (UC-II, 40mg/day) showing superior results to glucosamine+chondroitin in a head-to-head trial for knee OA, and collagen peptides demonstrating cartilage-regenerating effects in laboratory and animal studies. Limitations include fewer large-scale trials compared to glucosamine, study quality varying (many with small sample sizes), long-term effects less well documented, and the optimal type and dose not yet clearly established.

Dosage

Hydrolysed collagen: 10g daily (powder or liquid). UC-II (undenatured Type II): 40mg daily. Allow 3–6 months for potential effects.

Safety

Very well tolerated with few reported side effects. Mild digestive symptoms occasionally. Bovine collagen unsuitable for those avoiding beef products. Marine collagen unsuitable for fish allergies. No known significant drug interactions.

Head-to-Head Comparison

Effectiveness

Both supplements show modest benefits for OA symptoms. Glucosamine has more research behind it but results are inconsistent. Collagen has fewer but more consistently positive studies. The one direct comparison trial (UC-II vs glucosamine+chondroitin) favoured collagen. Neither reliably slows disease progression or rebuilds cartilage in established OA.

Speed of Effect

Glucosamine: 8–12 weeks. Collagen: 4–12 weeks (some studies report benefits as early as 4 weeks).

Cost (UK prices, approximate)

- Glucosamine sulphate 1,500mg: £8–£15/month
- Hydrolysed collagen 10g/day: £15–£30/month

- UC-II 40mg/day: £15–£25/month

Glucosamine is generally cheaper, though prices vary widely by brand and quality.

Convenience

Glucosamine: 1–3 capsules daily (easy). Collagen powder: needs to be mixed into drinks or food (slightly less convenient but tasteless). Collagen capsules: available but may require multiple capsules for full dose.

Can You Take Both?



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Yes. There are no known interactions between glucosamine and collagen, and some experts suggest they may work through complementary mechanisms. Some combined supplements are available. However, adding both increases cost, and the evidence for combination therapy

is limited — it may be more practical to try one for 3 months before considering adding the other.

Our Recommendation

- Try collagen first if you're new to joint supplements — the evidence is more consistently positive and it's very well tolerated
- Choose glucosamine sulphate if you prefer the more established option — ensure it's glucosamine sulphate (not hydrochloride)
- Give either supplement at least 3 months before judging effectiveness
- Remember — supplements complement but do not replace exercise, weight management and medical treatment

Neither glucosamine nor collagen is a magic bullet, but both may provide modest benefits for arthritis symptoms. The most important things for your joints remain exercise, maintaining a healthy weight and following your medical treatment plan. Our virtual assistant can help you decide which supplement might be right for your situation.

This article is for general information only. Consult your GP or pharmacist before starting supplements, especially if you take other medications.

Q&A

Question: What Is It?

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